

Cold Oxygen

Why a donor organ on ice bends the rules of oxygen delivery, and what that means for any carrier meant to help it

The name of the foundation that hosts this blog contains a technical term that most visitors read as a slogan. Cold ischemia is the interval during which an organ removed from a donor sits cooled and without blood flow, waiting to be placed in a recipient. It is measured in hours. It is one of the few variables in the whole process that a human being can shorten, and it is the reason that every organ is packed in ice and rushed in a cooler.

This essay is about the physics and biochemistry of that interval and about a question that follows from it. If oxygen carriers are to help a donor organ, they must work at temperatures far below the ones for which they were designed. Does a molecule tuned for the warmth of a living body behave sensibly at four degrees? The answer turns out to be interesting, partly counterintuitive and central to how one might build a preservation product around a BHOC.

What the cold buys and what it costs

Cooling an organ is a bargain with biochemistry. Metabolic reactions slow as temperature falls, following a rule of thumb that rates drop by a factor of roughly two to three for every ten degrees Celsius. By the time a kidney or liver sits at about four degrees, its energy demand is a small fraction of what it was at body temperature; figures around a tenth or less are commonly quoted, though the exact fraction differs among tissues and conditions. The organ is not dead. It is in a kind of suspended animation.

The bargain has a cost. Even at low demand, the cells still consume some energy, and in the absence of blood flow they have no oxygen to make it with. They burn through their stores of adenosine triphosphate, the molecule that powers pumps and repairs. The pumps that keep sodium and calcium in balance falter. Cells swell. Mitochondria, the small structures that make energy, begin to accumulate metabolic intermediates. The longer the interval, the more of this damage builds.

One of the most striking discoveries in this area concerns a molecule called succinate. During ischemia, mitochondria in many tissues accumulate succinate, a compound normally consumed in the energy-producing cycle. When blood flow returns and oxygen floods back, the built-up succinate is oxidized rapidly and drives a burst of reactive oxygen species from the mitochondria. That burst is thought to be a major driver of the injury that occurs on reperfusion, the moment the organ is reconnected. In other words, the damage is not simply from being without oxygen. It is from the way oxygen returns.

This changes the question. The problem is not only how to keep cells alive in the cold. It is how to prepare them for the moment of reconnection so that the return of oxygen is orderly rather than explosive.

The counterintuitive physics of oxygen in the cold

Now consider oxygen itself. Two physical facts collide at low temperature.

The first is that gases dissolve better in liquids when the liquid is cold. Oxygen is more soluble in water at four degrees than at thirty-seven, by a factor that is significant though not enormous. So a cold perfusion fluid can hold more dissolved oxygen at a given pressure. Combined with a lower metabolic demand, this means that the dissolved oxygen in a properly oxygenated cold solution may suffice, without any carrier, to supply an organ at rest.

The second is that hemoglobin binds oxygen more tightly in the cold. Cooling shifts the oxygen-binding curve to the left, lowering the P50, the pressure at which half the hemoglobin is saturated. The shift is substantial. A hemoglobin with a P50 in the thirties at body temperature has a much lower P50 at four degrees, so that it holds onto its oxygen until the surrounding pressure is very low. Physiologists know this effect from cold-water diving and from animals that let their extremities cool.

Put these two facts together and the picture is odd. In the cold, dissolved oxygen is more abundant and hemoglobin is more reluctant. A carrier that was designed to release oxygen at a moderate pressure in a warm body becomes, in ice, a molecule that grips its cargo. That is not necessarily fatal for the approach, because the organ needs so little oxygen and because the perfusate can be oxygenated to a very high pressure. But it means the role of a carrier in cold preservation is not a copy of its role in the body. It has to be reconsidered from first principles.

What the clinical evidence shows about oxygen alone

Before asking what a carrier adds, it helps to know how much oxygen alone accomplishes. Here the clinical evidence is unusually good, at least for the liver.

A randomized trial published in the New England Journal of Medicine in 2021 compared livers from donors after circulatory death that were preserved by static cold storage, the ice-cooler method, with livers that received a period of hypothermic oxygenated machine perfusion before implantation. The perfusion machine circulated an oxygenated solution through the liver at low temperature. Without going beyond what I could confirm, I recall that the trial found a substantially lower rate of symptomatic bile duct strictures in the perfused group, roughly 6 percent against roughly 18 percent. A careful reader should confirm those figures in the original report, but the direction and rough size of the effect are widely cited.

The important point is what the machine did not use. It did not circulate red cells. It did not use a hemoglobin carrier. The oxygen was dissolved in a cold, acellular fluid. That is a demonstration that in the cold, with a resting organ, dissolved oxygen can be enough to change outcomes. The proposed mechanism is that supplying oxygen during the cold interval lets mitochondria restore some of their energy stores and clear the succinate that would otherwise fuel the burst on reperfusion.

Any argument for a carrier in cold preservation must therefore start with humility. The baseline is not ice alone. The baseline is a machine that already delivers oxygen well, at least in the liver. The carrier would have to show a benefit beyond it.

Where a carrier might still add value

There are at least four situations in which a carrier could plausibly earn its place.

The first is static cold storage with an additive. Many organs, especially kidneys in many centers, travel in a cooler with no machine at all. Machines cost money, need staff and are not available at every donor hospital. A preservation solution that included an oxygen carrier could give a slow supply of oxygen to an organ sitting in a bag, without any pump. A developer of a lugworm-derived giant hemoglobin has pursued exactly this idea, and reported a first-in-human use in a kidney in France in 2016. The evidence base is small and largely from the developer, and I would treat it as suggestive.

The second is the warmer end of the range. Between the cold of four degrees and the warmth of body temperature lies a zone, roughly twenty to thirty degrees, that some groups use for what they call subnormothermic perfusion. At those temperatures the organ has substantial metabolic demand but dissolved oxygen alone falls short, and a carrier's affinity shift is smaller. This is the region where a carrier's properties fit the task best.

The third is the normothermic machine. At body temperature, machines that perfuse an organ for hours use red cells as the carrier, and those red cells must come from donated blood. A cell-free carrier that could replace them would remove a dependence on blood supply, ease logistics and, in principle, allow perfusion to be offered in places without blood banks. This is perhaps the most straightforward application, because the temperature matches the body, and the molecule is working in conditions it was designed for.

The fourth is the reperfusion moment. If a carrier could be present in the perfusate at the moment the organ is reconnected, it might buffer the return of oxygen. This idea is more speculative, but it matches the mechanism: the injury comes from a burst, and a carrier could in theory smooth it.

The advantage nobody mentions enough

One argument in favor of using a carrier in an isolated organ is powerful and deserves more attention than it gets. The history of oxygen carriers in the bloodstream is a history of systemic side effects: blood pressure elevation, effects on the heart, effects on the kidney, effects arising from the scavenging of nitric oxide in the vessel wall. Those effects are what stopped several programs.

An organ on a machine is outside the body. The carrier circulates through the organ and back to a reservoir. If the organ is flushed before it is connected to the recipient, most of the carrier is washed out, and the recipient receives little of it. In effect the drug is administered to the organ, not to the patient. The exposure that harmed trial subjects in the past is largely avoided.

I want to be careful. The claim is that systemic exposure is reduced, not eliminated. Some carrier remains in the small vessels and tissue, and its fate after reperfusion needs to be studied. The organ's own vessel lining, exposed to a hemoglobin solution, may show local effects. But as a way of reintroducing an oxygen carrier to medicine while limiting its risk, the ex vivo setting is one of the most sensible I know. Ask the right question of any program: how

much carrier remains in the organ after flush, how was it measured, and what happened to the recipients?

A day in the life of a kidney

Abstractions are easier to judge against a concrete timeline. The following sequence is a composite for illustration, not a record of any real case.

The donor is declared, and the kidneys are recovered in an operating room while the donor is still on support. Cold preservation fluid is flushed through the vessels until the organ blanches, then the kidney is trimmed, bagged in sterile layers and packed in a cooler with ice. From that moment the clock of cold ischemia is running.

The kidney travels. It may go by road to an airport, by air to another state, by road again to a hospital. Each handoff is a chance for delay: a missed connection, a weather hold, a courier who cannot find the loading dock. Meanwhile, at the receiving hospital, a recipient is called in, tested and prepared, and a surgical team is assembled. If the recipient turns out to be unsuitable at the last minute, the kidney is redirected, and the clock keeps running.

When the kidney arrives it is inspected, and someone decides whether it looks well enough to use. That decision, made under time pressure, on appearance and a few numbers, is a place where a lot of quality and a lot of anxiety meet. Then comes the surgery, during which the kidney is warmed by the body and stitched into place, and finally the clamps come off and blood flows. That moment of reconnection, the reperfusion described earlier, is when the injury from the cold interval expresses itself.

Nothing in that sequence gives the organ oxygen. If a preservation system could add even a modest supply of oxygen during the transport, in a cooler with no pump, it would touch every stage of the journey without asking the couriers to do anything differently. That is why the static cold storage additive is, to my eye, the most attractive application of a carrier, even though the evidence is thinnest there.

Kidneys are not livers

A last caution concerns generalizing from the liver trial. Organs differ in their metabolism, their vascular architecture and their sensitivity to cold. The liver is a metabolically busy organ with a dual blood supply, and its injury pattern after preservation involves the bile ducts, which are exquisitely sensitive to oxygen deprivation. The kidney has a different profile. Its tubules consume large amounts of energy to reabsorb salts, and its characteristic post-preservation problem is delayed function, when the organ is slow to begin working. Machines for kidneys have been studied for years, and randomized trials have found reductions in delayed function with hypothermic machine perfusion compared with ice storage, although the size of the benefit varies among studies and the machines used are usually not oxygenated.

The upshot is that a carrier for kidneys would be entering a field with a well-established machine comparator, in which the addition of oxygen itself is still being tested. A prudent developer would design a kidney study with both questions in view: does oxygen help, and does a carrier add anything to oxygen delivered by simpler means? An advocate should want both

questions asked, because a positive answer to the second would be far more persuasive than a positive answer to the first alone.

The instrument problem, again

A cold perfusion carrier raises measurement problems that echo an earlier post. Clinicians assess a machine-perfused organ by looking at it. Color, texture and the way it blanches and refills are used as signs of quality, and an experienced surgeon judges viability partly by eye. A solution full of red-brown hemoglobin changes the look of the organ and may change those judgments. Perfusate chemistry tests, such as lactate or enzymes released from damaged cells, are used to judge injury, and a pigment can interfere with colorimetric assays.

The remedy is not to avoid carriers but to validate the assessments in their presence, and to agree on markers that do not depend on color. This is the sort of unglamorous protocol work that decides whether a technology is adopted.

The choice of molecule

If I were advising a developer about preservation in the cold, I would ask which molecule and why.

A carrier with a very high affinity would be a poor unloader at four degrees. A carrier with a moderate affinity at body temperature may become high-affinity in the cold, which is acceptable if the organ needs little oxygen but is wasteful if the goal is delivery. A carrier with a large size, such as a giant worm hemoglobin, stays in the vessels and may show good antioxidant behavior, which suits reperfusion. A bovine polymer sits between. It has a long track record of manufacturing and human use, a middling affinity and a modest size distribution.

There is also the option of a cold-tuned design, a carrier engineered so that its affinity at low temperature is closer to what the organ needs. Whether such a design is feasible is an open question. I raise it because the cold has been treated as a constraint to be endured rather than a design parameter to be used.

The choice of comparator deserves a sentence of its own. A carrier study that compares against nothing but ice will look better than one that compares against a simple oxygenated machine, and sponsors will be tempted by the easier comparison. Patients are better served by the harder one, because the harder one tells them whether the carrier is worth the added complexity, cost and regulatory burden.

Why this matters to patients

For a patient waiting for a donor organ, the cold interval is invisible. It happens to somebody else's organ, in a cooler, on a road or a runway, and is measured in hours nobody sees. Yet its length and quality shape how long the organ works after it arrives, and whether it works at all. A meaningful share of organs that are recovered are never used, for reasons that include concern about quality. Improved preservation is one of the few interventions that could widen the supply without asking a single additional person to donate.

That is the reason a foundation devoted to the cold interval should follow oxygen carriers closely and demand clear evidence. It should also be a wary audience. The strongest current data show that oxygen, delivered by a machine in a simple fluid, already helps. The carrier's role is to extend that benefit to places without machines, to warmer perfusion and to the moment of reconnection. Those are worthwhile goals, but each requires its own evidence.

What a good study would look like

To close, imagine the study I would want to see. It would take organs from donors of comparable type, randomize them to standard preservation or to preservation with a defined carrier, and record the time in the cold. It would measure the carrier remaining in the organ before implantation. It would follow recipients for function of the organ, injury markers, delayed function, complications and survival of the graft at one year. It would report all results, positive and negative, and it would be overseen by an independent committee.

It would not be a large study at first. A small, careful, transparent study of that kind would be worth more than a dozen glowing case reports. And it would be a study a patient community could support without hesitation, because it asks a modest question: does adding oxygen to the cold help?

Further reading

The randomized trial of hypothermic machine perfusion in liver preservation appeared in the New England Journal of Medicine in 2021, and follow-up and review articles discuss the mechanism. The role of succinate accumulation and reperfusion injury was described in work by Chouchani and colleagues in 2014. The developer of a lugworm-derived carrier has published descriptions of its preservation use. Quantitative claims about temperature effects on metabolism and hemoglobin affinity are approximate and should be checked in physiology references.